13 HYDROGRAPHY

FLOW MEASUREMENT

1. Which of the following discharge rates and periods of time will most nearly accumulate 1 acre-foot?

(A) 25 gal/min for 24 hr
(B) 1 ft^3/sec for 12 hr
(C) 30 miner's in for 6 hr
(D) 50 gal/min for 8 hr

Solution: An acre-foot is the quantity of water required to cover an acre to a depth of 1 ft, or 43,560 ft^3. A flow of 1 ft^3/sec over 12 hr, or 43,200 sec, comes closest to accumulating 1 acre-foot.

Answer is (B)

2. What is the meaning of the word *stage* as it is used in hydrography of streams?

(A) The stage of a stream refers to the amount of water flowing past a given section at a given time.
(B) The stage of a stream refers to the velocity of the water flowing past a given section at a given time.
(C) The stage is the height of the water surface above a given datum.
(D) The stage refers to the location of a gauging station on a stream bank.

Solution: The stage of the water in a stream is its height above a given datum.

Answer is (C)

3. Of what use is a stage-discharge curve for a stream?

(A) A stage-discharge curve is an indispensable tool for predicting the occurrence of floods.
(B) The computation of a stage-discharge curve is not necessary for streams with a steady flow.
(C) A stage-discharge curve correlates the height of the water and the flow of a stream.
(D) A stage-discharge curve correlates the velocity of the water and discharge of a stream.

Solution: The primary use of a stage-discharge curve for a particular stream at a particular section is to estimate the flow of the water by measuring the stage, or height, of the water surface.

Answer is (C)

4. Which of the following is not used to measure the level of the water in a stream?

(A) staff gauge
(B) hook gauge
(C) wire weight gauge
(D) vane gauge

Solution: Each of the gauges listed is used to measure the stage of water in streams, except for the vane gauge.

Answer is (D)

5. Which of the following instruments is equipped with a mercury manometer assembly?

(A) an optical current meter
(B) a Haskell meter
(C) a Parshall flume
(D) a bubble gauge

Solution: A manometer is an instrument used to measure the pressure of the fluid in a tube, and is an integral part of a bubble gauge. Nitrogen gas is discharged through a tube at a fixed elevation in a stream below its minimum stage, and the pressure at the orifice is measured by the manometer. The pressure in the tube varies directly with the height of the water.

Answer is (D)

6. What government agency is responsible for compiling stream flow data in the U.S.?

(A) the Bureau of Reclamation
(B) the National Geodetic Survey
(C) the National Ocean Service
(D) the U.S. Geological Survey

Solution: The Water Resources Division of the U.S. Geological Survey publishes an annual series of water supply papers concerning the surface water in the U.S. Data on stream flow may be obtained from the district offices of the U.S. Geological Survey.

Answer is (D)

7. The discharge of a stream is the amount of water, in cubic feet per second, flowing past a given section in a given time. The discharge is the product of which of the following factors?

(A) the pulsation of the stream and the depth of the main channel
(B) the cross-sectional area and the mean velocity of the water
(C) the datum of the stage measurements and the average staff gauge readings
(D) the pulsation of the stream and the mean velocity of the water

Solution: The discharge of a stream is the product of the cross-sectional area at right angles to the flow and the mean velocity of the water.

Answer is (B)

8. Which of the following is not a factor in choosing a gauging site along a river?

(A) A gauging site should be chosen where very low flows may be successfully recorded.
(B) A gauging site should be chosen where there are high stable banks clear of brush and where there is a uniform bottom.
(C) A gauging site should be chosen where there is a natural constriction providing relatively constant stage and discharge readings.
(D) A gauging site should be chosen on a gradually curving section of the river near a confluence.

Solution: A gauging site near a confluence or on a curve in the river would invite variable readings in the stage measurements. The location of a gauging site should be on a straight section of the river far from any confluence.

Answer is (D)

9. Which sort of gauge is incremented after being placed in position?

(A) staff gauge
(B) float gauge
(C) wire weight gauge
(D) inclined staff gauge

Solution: An inclined staff gauge is usually graduated after being set in a permanent foundation since it follows the slope of the bank.

Answer is (D)

10. Which of the following current meters is most frequently used in the U.S.?

(A) Ott current meter
(B) Hoff current meter
(C) Price current meter
(D) Haskell current meter

Solution: The Price current meter, developed by the U.S. Geological Survey, is usually the meter of choice in the U.S. It has the advantages of being useful over a wide range of velocities, being repairable in the field, and having bearings that stand up well to silt.

Answer is (C)

11. The Price current meter is equipped with a buzzer or sounder that may be set differently depending on the conditions. Which of the following statements best characterizes the available settings?

(A) The electrical contact may be set to sound with every revolution or every other revolution, depending on the velocity of the water.

(B) The meter may be set to emanate a mechanical click with every revolution in slow currents, or an electrical buzz in fast water.

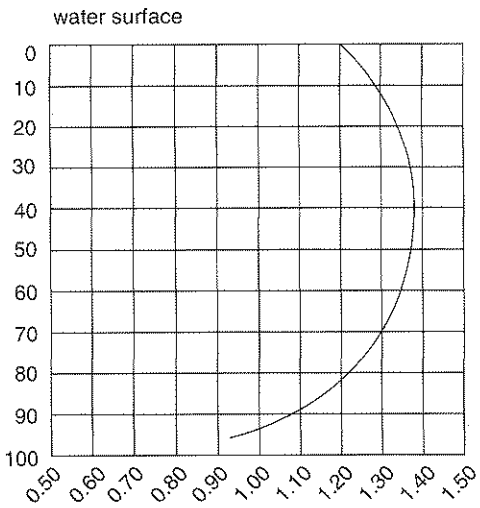
(C) The meter may be set to sound with every revolution in slow currents or with every fifth revolution in fast water.

(D) The faster the current, the more frequently the meter sounds, sounding with every third revolution in slow currents and every revolution in faster water.

Solution: The pentacount setting on the Price current meter causes the instrument to sound with every fifth revolution and is used in water with a current exceeding 6 ft/sec.

Answer is (C)

The following four problems refer to the illustration shown.



12. If the arc in the illustration is considered to be a typical vertical-velocity curve, what units are being expressed by the numbers across the bottom of the chart?

(A) the depth of the water in tens of feet
(B) the velocity of the water in feet per second
(C) the quantity of water in cubic feet
(D) the quantity of water in miner's inches

Solution: The typical arrangement of a vertical-velocity curve is to express the depth of each reading as a percentage of the distance from the water surface to the bottom on the ordinate. The velocity of the water is expressed in feet per second along the abscissa.

Answer is (B)

13. Which of the following shapes does the vertical-velocity curve most closely approximate?

(A) a parabola
(B) a cycloid
(C) a helicoid
(D) a circular curve

Solution: The vertical-velocity curve most closely resembles a parabola, and for most calculations is assumed to be one.

Answer is (A)

14. Which of the following elements usually affects the shape of a vertical-velocity curve?

(A) the direction of the wind
(B) whether the water is covered with ice or not
(C) the distance from the banks
(D) all of the above

Solution: Each of the conditions listed has some influence over the shape of the vertical-velocity curve for a particular section. Another element that could be included would be the depth of the channel.

Answer is (D)

15. Which of the following methods would yield the most accurate indication of the mean velocity expressed by the vertical-velocity curve?

(A) the two-point method
(B) the subsurface method
(C) the six-tenths method
(D) division of the area between the curve and the ordinate by the length of the ordinate

Solution: The assumption that the vertical-velocity curve is a parabola allows for several methods of approximation of the mean velocity expressed by a particular curve. The most accurate method is the most time-consuming and can be summarized as finding the axis of the parabola. Dividing the area between the curve and the ordinate by the length of the ordinate yields the length of a particular axis, which will usually occur between 10 and 30% of the total depth.

Answer is (D)

16. Which of the following statements best describes the vertical integration method of current measurement?

(A) The meter is placed at 20% of the total depth of the stream, and the mean velocity is calculated with a previously established coefficient.

(B) The meter is placed at 60% of the total depth of the stream, and the observed value is taken as the current's mean velocity.

(C) The meter is lowered at a slow and uniform rate to the bottom of the stream and then raised at the same speed. The number of rotations of the meter is recorded and the mean velocity found by using the meter's rating table.

(D) The meter is lowered to 50% of the total depth of the stream in the same position at several different times. The mean velocity is taken as the average of all readings.

Solution: A horizontal-axis current meter is preferred in the integration method of flow measurement; a vertical-axis current meter should not be used. The integration method is based on the assumption that all horizontal and vertical current forces have equal influence on the current meter.

Answer is (C)

17. Which of the following characteristics of a vertical-velocity curve would be indicative of the flow pattern created in an ice-covered stream?

(A) The shape of the curve would more closely approximate a circular curve than a parabolic curve.

(B) The curvature would be sharper near the surface of the water.

(C) The mean velocity would occur near 75% of the total depth instead of 60%.

(D) Fewer observations would be necessary due to the flatness of the curve.

Solution: Due to the friction created by the underside of the ice, the velocity of the current is significantly decreased near the water surface, giving the vertical-velocity curve a sharper curvature near the top.

Answer is (B)

18. When calculating the average velocity represented by a vertical-velocity curve using the two-point method, at what depths should readings be taken?

(A) 10% and 90% of the total depth
(B) 20% and 80% of the total depth
(C) 30% and 70% of the total depth
(D) 40% and 60% of the total depth

Solution: The average of readings taken at 20% and at 80% of the depth from the water surface will closely approximate the average velocity represented by a vertical-velocity curve. This method is widely used and yields good results in most cases. It also avoids the time-consuming process of finding the average area included within a particular vertical-velocity curve.

Answer is (B)

19. Which of the following circumstances makes the application of the two-point method inappropriate for finding the mean current velocity of a stream section?

(A) a stream that is less than 2 ft deep
(B) high current velocities
(C) the use of a horizontal-axis current meter
(D) both A and B

Solution: The use of a horizontal-axis current meter, such as a Hoff meter, does not prevent the application of the two-point method. However, very high current velocities, depths of less than 2 ft, and high levels of debris in the water can seriously degrade the results.

Answer is (D)

20. What does it mean to rate a current meter?

(A) A current meter is rated by the maximum and minimum current velocities it can measure.

(B) A current meter is rated according to its tolerance to various conditions of silt, current velocities, and methods of suspension.

(C) A current meter is rated to correlate its number of revolutions per unit of time to specific current velocities.

(D) A current meter is rated as mechanical, optical, or electrical.

Solution: A current meter is rated by towing it through still water at various speeds while counting its revolutions. This is necessary because each meter has slightly different bearing frictions and other physical attributes that affect the rate at which it revolves.

Answer is (C)

21. Which of the following statements describes a typical purpose of a weir?

(A) A weir is used to stabilize a sounding line.

(B) A weir is a type of station-rating curve.

(C) A weir is a control structure used in canals, shallow streams, and sewers.

(D) A weir is a small portable flume with a throat of 12 in or less.

Solution: A weir is built much like a dam; however, water is allowed to flow over a weir, usually through a notch. A weir is often used to measure flow.

Answer is (C)

22. What sort of weir is used primarily to measure water flow?

(A) a rectangular weir
(B) a trapezoidal weir
(C) a triangular weir
(D) a sharp-crested weir

Solution: A broad-crested weir may function as a water-measuring structure, but that is usually not its primary purpose. However, the exclusive function of a sharp-crested weir is water measurement.

Answer is (D)

23. What does the term *head* indicate when referring to a weir?

(A) the width of the crest of the weir
(B) the cross-sectional shape of the wier
(C) the width of the channel approaching a weir
(D) the depth of the flow over a weir

Solution: The head is the height of the water that flows over the crest of the weir. The head is usually measured from the top of the crest to the surface of the water pooled upstream of the weir. In calculating the discharge, the measurement is made at a distance that is three times the length of the head upstream from the crest of the weir and is then corrected for the velocity of the approaching current where the channel is contracted.

Answer is (D)

24. Neglecting the approach velocity of the flow into a 90° sharp-crested V-notch weir, if the head is 6 in, what is the discharge in cubic feet per second?

(A) 1.22 ft^3/sec
(B) 1.03 ft^3/sec
(C) 0.83 ft^3/sec
(D) 0.45 ft^3/sec

Solution: The general formula for the calculation of the discharge over a 90° triangular weir, neglecting the velocity, which is generally quite low with such control, is

$$Q = 2.49H^{2.48}$$

Q is the discharge in cubic feet per second, and H is the height of the head in feet.

$$\begin{aligned} Q &= (2.49)(0.5\text{ ft})^{2.48} \\ &= (2.49)(0.1792\text{ ft}) \\ &= 0.45\text{ ft}^3/\text{sec} \end{aligned}$$

Answer is (D)

25. What sort of gauge is most often used to measure the head of water flowing over a weir?

(A) a wire weight gauge
(B) a crest-stage gauge
(C) an inclined staff gauge
(D) a hook gauge

Solution: A hook gauge is most often used to measure the head of a flow over a weir. The gauge is mounted upstream, at a distance from the crest that is three times the head, with the point of the hook coincident with the water surface.

Answer is (D)

HYDROGRAPHY AT SEA

26. What is a bathymetric survey?

(A) a survey conducted by sampling material on the seabed
(B) a survey of the depth of water
(C) a survey conducted by sampling seawater
(D) a survey of the salinity of seawater

Solution: A bathymetric survey involves the measurement of the depth of oceans and other large bodies of water. The term may also refer to the mapping of the topography of the seabed based on soundings.

Answer is (B)

27. Mapping the seabed by echo sounding involves the transmission of acoustic waves through seawater. What is the most serious source of error in this procedure?

(A) the width of the acoustic beam
(B) the angle of incidence at the target
(C) spurious echoes
(D) the velocity of the signal

Solution: The velocity of the acoustic signal through seawater is the most important source of error in echo sounding.

Answer is (D)

28. Which of the following affects the velocity of sound through seawater?

(A) salinity
(B) temperature
(C) density
(D) all of the above

Solution: The density of the water has a large impact on the speed of the acoustic signal. The temperature, the salinity, and the density vary with the depth, but like the atmospheric corrections applied to the signal from an EDM, these elements are difficult to quantify.

Answer is (D)

29. A depth determined by echo sounding may be checked with a thermometric depth. Which of the following best describes a thermometric depth?

(A) a probe giving continuous telemetry on salinity, temperature, and sound velocity lowered through the water column
(B) the isothermal determination of the steric anomaly at 1 atmosphere
(C) the depth determined by lowering a thermometer subject to hydrostatic pressure along with a thermometer protected from hydrostatic pressure
(D) the depth calculated by comparing the temperature from a thermistor at the surface with a similar instrument lowered at depth

Solution: The measurement of the temperature-pressure gradient with a pair of thermometers, one protected from and one exposed to hydrostatic pressure, can determine depth within 5 m.

Answer is (C)

30. Which of the following devices involved in echo sounding can be categorized as transducers?

(A) projectors
(B) hydrophones
(C) platen
(D) both A and B

Solution: An instrument that transforms one kind of energy to another is a transducer. The projectors in echo-sounding equipment are driven by electrical power, but produce an acoustic signal. The hydrophones receive that signal after it has been reflected and transform it back into an electrical impulse.

Answer is (D)

31. What is the frequency range usually employed in echo sounding?

(A) 1200–1500 MHz
(B) 100–300 MHz
(C) 1500–3000 KHz
(D) 1–300 KHz

Solution: The frequency range most often used by echo-sounding equipment is 1–300 KHz.

Answer is (D)

32. Which of the following affects the detection capabilities of echo-sounding equipment?

(A) the nature of the target and the angle of incidence of the wavefront with respect to the target
(B) the pulse length, duration, and the beam width of the transmission
(C) the recording medium
(D) all of the above

Solution: Resolution is usually limited to half of the pulse length, unless the beam incidence is not normal to the target. The recording medium may not have the capability of displaying the echoes received if resolution is necessary in real time.

Answer is (D)

33. As the frequency of an acoustic wave is increased, what are the attendant effects for a given transducer size?

(A) The wavelength is shortened, and the beam width is widened.
(B) The wavelength is lengthened, and the beam width is widened.
(C) The wavelength is lengthened, and the beam width is narrowed.
(D) The wavelength is shortened, and the beam width is narrowed.

Solution: For a given transducer size, as the frequency increases, the wavelength is shortened and the beam width is narrowed.

Answer is (D)

34. What is the relationship between the speed of sound through the atmosphere and the speed of sound through seawater?

(A) The speed of sound through the atmosphere is nearly four-and-one-half times slower that it is in seawater.
(B) The speed of sound through the atmosphere is nearly two times slower that it is in seawater.
(C) The speed of sound through the atmosphere is nearly the same as it is in seawater.
(D) The speed of sound through the atmosphere is nearly three times faster than it is in seawater.

Solution: The speed of sound through the atmosphere at sea level is approximately 1087 ft/sec. The speed of sound through seawater is approximately 4900 ft/sec, or nearly four-and-one-half times faster.

Answer is (A)

35. Which of the following statements best describes a bar check of echo-sounding equipment?

(A) The echo sounder is lowered with a small metal plate set at a known distance below the transducer. The sounder is then adjusted to read the calibration distance at each depth.

(B) A metal target is lowered on marked lines below the echo sounder. As the target is lowered to various depths, the echo sounder is adjusted to read the correct distance to the target.

(C) The stylus speed on the recorder is adjusted to correspond with the speed of the pulse from the echo sounder to the seafloor and back.

(D) The motor speed of the recorder is adjusted to correspond with the speed of the vessel in relation to the seafloor.

Solution: A bar check involves lowering a metal bar or disk to various depths while the echo sounder is adjusted to correctly record the known depths to the target.

Answer is (B)

36. Obstructions to navigation are sometimes so small as to escape soundings. Which of the following devices is designed to detect such hazards?

(A) backscatter meter

(B) vibro-corer

(C) wire sweep

(D) sediment dredge

Solution: The wire sweep, or drag, is used to locate pinnacle rocks, wrecks, and other hazards in channels that have been selected as safe routes at particular depths. The drag consists of wires or rods held up by buoys, which are then towed by two boats at the required depth. When an obstruction is encountered, the buoys describe two lines that converge at the hazard.

Answer is (C)

37. Which of the following are examples of seabed samplers?

(A) the reversing bottle, the swallow float, and the dredge
(B) the grab sampler, the corer, and the dredge
(C) the sheave, the sweep, and the corer
(D) the pocket scow, the T-bar, and the coaming

Solution: The corer is used to obtain a sample from several feet into the seabed, the dredge is designed to bring up loose material lying on top of the seabed, and the grab sampler penetrates the seabed.

Answer is (B)

38. Which of the following statements is not true of modern recording current meters used in measuring the flow of seawater?

(A) The direction of the flow is determined by an on-board magnetic compass.
(B) The rotor is protected from debris, and the instrument body is watertight.
(C) The recording current meter may be left unattended for long periods.
(D) The recording current meter is designed for use at several depths.

Solution: Most current meters are designed for use at specific depths. Several different types of meters may be arrayed simultaneously to determine flow at various depths.

Answer is (D)

39. What is a messenger as the term is used in hydrographic instrumentation?

(A) A messenger is a drogue chute with a sinker used to retard the movement of a float-type current meter.

(B) A messenger is an electronic meter that measures the salinity of seawater by its ability to transmit visible light.

(C) A messenger is a weight that falls down a line and is used to trigger instruments.

(D) A messenger is an electronic interface used to monitor a direct-reading current meter.

Solution: A messenger is a small weight with a slot that allows it to be attached to a line holding a hydrographic instrument at depth. At the appropriate moment, the weight is allowed to fall down the line to trigger the instrument.

Answer is (C)

HYDROGRAPHIC MAPS

40. When soundings are shown on a hydrographic map, what is the exact position of the measurement?

(A) The position of the sounding is represented by a + sign.

(B) The position of the sounding is represented by the decimal point.

(C) Every sounding on the map is connected with contour lines.

(D) The position of the sounding is represented by the letter *s*.

Solution: The position of the decimal point indicates the point at which the sounding was made.

Answer is (B)

41. Which of the following dimensions is equal to 1 fathom?

(A) 1 yd
(B) 6 ft
(C) 1 rod
(D) 4 ft

Solution: A fathom is equal to 6 ft.

Answer is (B)

42. What should distinguish the low-water line on a hydrographic map?

(A) The low-water line is the heaviest line on the map.
(B) The low-water line is indicated by a dashed line.
(C) The low-water line is the second heaviest line on the map.
(D) The low-water line is indicated by a dash-dot line.

Solution: The heaviest line on a hydrographic map is the high-water line. The second heaviest line is the low-water line.

Answer is (C)

43. Where the datum of a hydrographic map is mean low water, how may soundings that are below the datum be easily distinguished from those that are above?

(A) Soundings above the datum are usually circled, while those below the datum are not.
(B) Soundings that are below the datum are usually circled, while those above the datum are not.
(C) Soundings above the datum are usually expressed to the nearest foot, while those below the datum are expressed to the nearest tenth of a foot.
(D) Soundings below the datum are usually in black, while those above the datum are in some other color.

Solution: The usual practice is to show soundings below mean low water in black, while soundings above the datum are lettered in some other color.

Answer is (D)

44. The standard hydrographic symbols used by the USC&GS (now the National Geodetic Survey) in the preparation of nautical charts include which of the following?

(A) Land areas are in yellow, while marsh or beach areas are in green.
(B) Contour lines on land are shown in red, while underwater contours are in black.
(C) Depths of water are shown in fathoms, while elevations on land are shown in feet.
(D) The contour interval is 10 ft on the land but 5 ft under the water.

Solution: The usual convention is to show marshes and beaches in green and the upland areas in yellow.

Answer is (A)

45. The most convenient method of plotting soundings on a map is dependent on the procedures used to locate them in the field. Which of the following plotting techniques is best suited to soundings located by resection—that is, measuring two angles between three known points from the survey vessel?

(A) polar protraction
(B) the use of two tangent protractors
(C) the use of a three-armed protractor
(D) the tracing-cloth method

Solution: A three-armed protractor can be used to set off the angles measured at the point of the sounding and then moved across the plan until the three arms coincide with the three control stations. However, this technique cannot provide a unique solution when the three control points are on the circumference of a circle that also includes the position of the sounding.

Answer is (C)